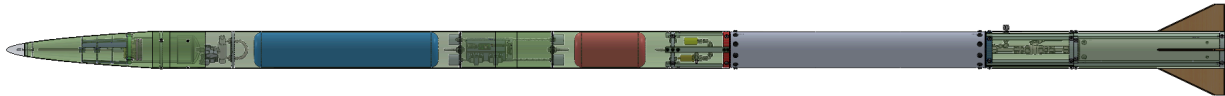


Motor Casing Design

This motor casing will need to withstand a maximum chamber pressure of 590 psi with a safety factor of 3 for NYU's first ever hybrid rocket.



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Project Specifications

The objective of this project is to design and manufacture a motor casing that will be able to withstand a maximum chamber pressure of 590 psi and a nominal burn time of 3.001 seconds. Initial propulsive parameters are derived from a simulation in Hybrid Rocket Analysis Program (HRAP) and a safety factor of 3 will be used.

Initial Design Parameters

The design solution will be informed by the following parameters and constraints:

- Affordable to manufacture (total cost < \$300)
- Easy to source (ex. McMaster Carr)
- Easy to manufacture using Makerspace and Maker Garage facilities
- Peak chamber pressure of 590 psi
- Average chamber pressure of 225 psi
- Burn time of 3.001 seconds
- Peak internal temperature of 3200 K
- Grain OD of 3.27 inches
- Grain length of 12 inches
- Maximum OD of 5.15 inches

Chamber Design

The construction of the thrust chamber will primarily consist of a length of round tube with an injector and nozzle on opposing ends affixed radial bolts. The fuel grain, along with a phenolic liner, will be held inside of the round tube section. A labeled cross-section of the construction can be seen in Figure 1.

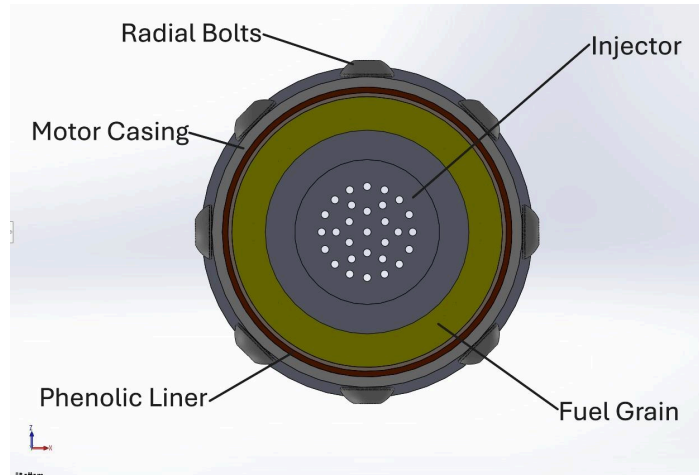


Figure 1a: Thrust Chamber Cross-Section

The fuel grain, made of paraffin wax, was initially designed to be 12 inches long with a 3.38 inch outer diameter and 1.57 inch inner diameter. This was done with the help of the Hybrid Rocket Analysis Program (HRAP) in order to optimize the motor's burn characteristics. However, in order to accommodate the use of a phenolic liner, the outer diameter will be decreased to 3.27 inches. This will slightly decrease the overall mass of the assembly but worsen its thermal insulation.

The phenolic liner was originally meant to be a canvas-cloth reinforced phenolic tube with an OD and ID of 3.625 and 3.27 inches, respectively. However, due to ordering and supply chain constraints, the liner was instead manufactured in-house using a fiberglass composite. Its primary purpose is to prevent hotspots and/or burnthrough by adding a layer of thermal insulation between the fuel grain and the outer casing. It also has the added benefit of preventing the build-up of burnt fuel on the inner wall of the motor casing. While the fiberglass does not perform as well as a true phenolic liner, in-house production means that the dimensions can be customized to fit the exact specifications of existing hardware. Previous designs utilized a layer of RTV silicone in between the motor casing and liner to fill the difference between their ID and OD, respectively, which is no longer necessary with the fiberglass.

The motor casing itself will be made out of $\frac{1}{8}$ inch thick aluminum round tube with a nominal OD of 3.75 inches. This will be able to withstand the required pressures during the burn and also more readily dissipate heat than a stainless

steel alternative. The higher strength-to-weight ratio of aluminum compared to steel is also a benefit in this weight-sensitive application.

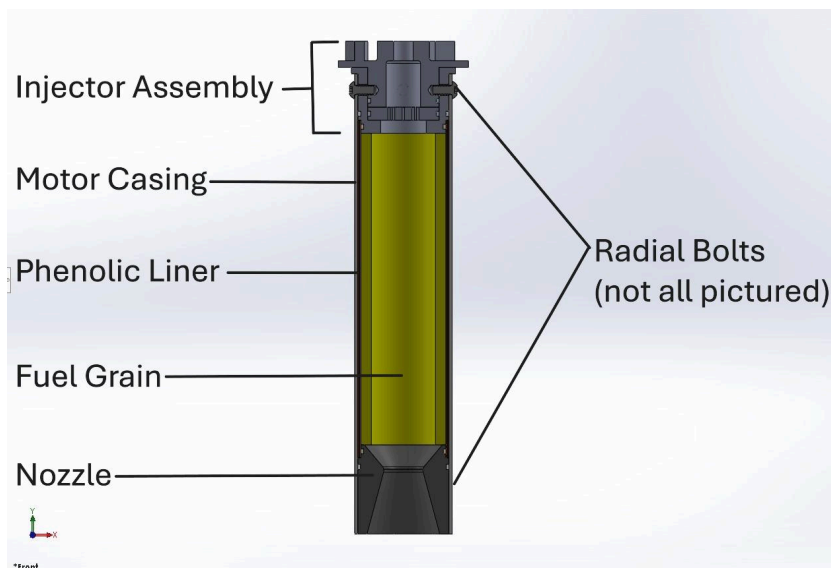


Fig 1b: Thrust Chamber Cross-Section

The nozzle is based on the design put forth by Matt Ricchiuti in April of 2025. It features an exit half-angle of 15 degrees and an expansion ratio of 3.55. It will be made via a combination of manual and CNC turning/milling out of a combination of an aluminum carrier and graphite throat.

The injector plate will be machined out of a disc of solid aluminum stock with a geometry designed by Jonathan Shan and Darren Lin. A port with a tapered thread will be included on the outward-facing face to connect with necessary plumbing. Ports for pressure transducers are also being considered for data collection purposes. The material may be changed pending further research into existing solutions and thermal estimations.

High-temperature silicone O-rings ($\frac{1}{8}$ inch fractional width, dash 236) will be used to provide a pressure seal within the combustion chamber. The sizing for these O-rings was done following Half Cat Rocketry's HCR-5100 Guidebook published on their liquid rocket, Mojave Sphinx, as well as Parker's O-Ring Handbook.

Both the nozzle and injector plate will be secured with 8 $\frac{3}{8}$ -16 radial bolts. This method was chosen over an axial tie rod system, inspired by Half-Cat's

Mojave Sphinx, to provide improved mounting clearance for the aerostructures team and eliminate any concerns over flange bending and/or failure.



Fig 2: Axial Tie Rod Design

Pressure Calculations

While rough hand calculations were performed in the early stages of conceptual development, a bulk of the final calculations were performed in Excel. The values in question include hoop stress, axial stress, bolt shear, bolt tear-out, etc. The equations and final values can be seen in Figure 3 below. Further data and values can be found in the spreadsheet linked in Appendix B. Finally, a simple pressure simulation ($P = 725$ psi) in ANSYS was used to validate calculations and identify notable stress concentrations.

Stress Calculations			
Hoop Stress			
sigma_h	5145 psi		$\sigma_h = \frac{P \times r}{t}$
FOS	6.8		
Axial Stress			
sigma_a	2572.5 psi		$\sigma_a = \frac{P \cdot D}{4t}$
FOS	13.6		
Casing Tensile Stress			
sigma_tensile	2968 psi		$\sigma_{tensile} = \frac{\frac{\pi}{4} D_{i,casing}^2 \times MEOP}{[(D_{o,casing} - t)\pi - N \times d_{bolt,major}] \times t}$
FOS	11.8		
Bolt Shear			
sigma_shear	14172 psi		$\sigma_{bolt\ shear} = (\frac{\pi}{4} D_{i,casing}^2 \times MEOP) / (N \times \frac{\pi}{4} d_{bolt,minor}^2)$
FOS	6.8		
Bolt Tear-Out			
F_Bolt	1179 psi	$F_{bolt} = \frac{\frac{\pi}{4} D_{i,casing}^2 \times MEOP}{N}$	$\sigma_{tear-out} = \frac{F_{bolt}}{E_{min} \times 2t}$
sigma_shear	5527 psi		
FOS	5.8		
Bearing Stress			
sigma_bearing	12572 lbf	$\sigma_{bearing} = \frac{F_{bolt}}{d_{bolt,major} \times t}$	
FOS	4.8		

Fig 3: Pressure Calculations

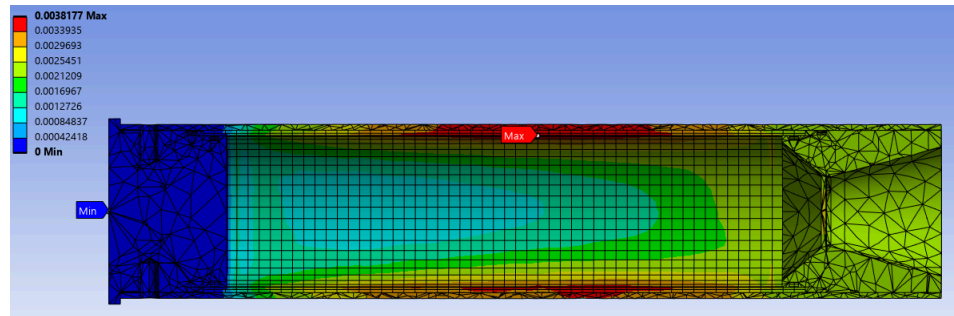


Fig 4a: ANSYS Deformation Results

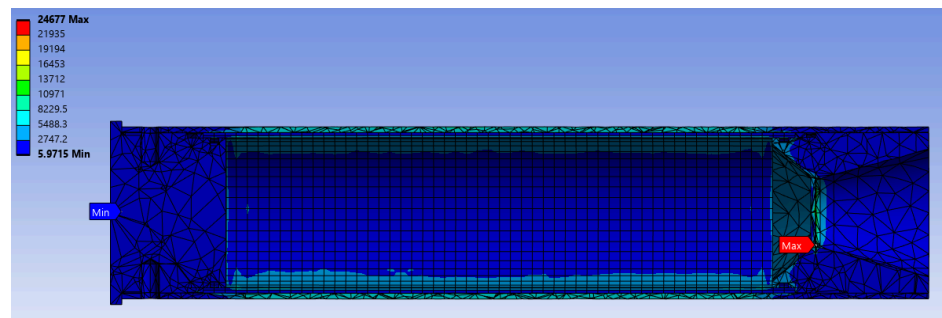


Fig 4b: ANSYS Stress Results

Temperature Simulations

Simplified 1-D thermal estimates were initially used to approximate the temperature of the motor casing during the estimated length of the burn. Similar to the pressure calculations, ANSYS was again used to validate the calculated numbers; however, this time the simulated results were relied on more heavily given the complex nature of transient thermal problems.

According to the combustion values provided by HRAP, an internal ambient temperature of 3200K was applied. This assumes a perfectly efficient combustion process, and along with an external ambient temperature of 30 C and perfectly insulated conditions on the ends, makes for a conservative estimate that leaves headroom for imperfections and inefficiencies throughout the system. The wax thickness was set to its estimated value at the end of the burn to apply a worst-case scenario. Finally, a convection coefficient of $1000 \text{ W/m}^2\text{-K}$ was applied to the inner face to approximate the gas flowing during the burn and the simulation time set to three seconds.

Figure 5 displays the simulated results. The values were found to be well within acceptable margins and the motor casing is not at risk of losing significant material strength due to softening.

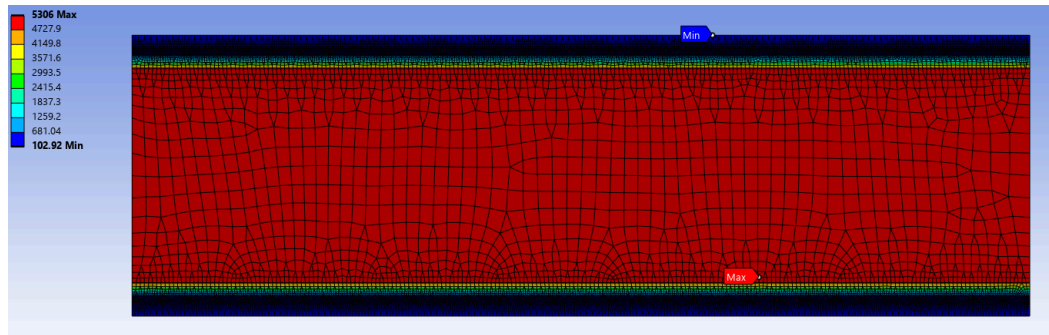


Fig 5: Ansys Thermal Results

Safety

In order to verify that the Motor Casing and its associated hardware have been adequately designed and manufactured, they will undergo a hydrostatic and cold flow test before a full-scale static fire. This will be done to ensure that any structural failures occur in a safe testing environment without risk of injury or destruction of property. The hydrostatic test will be performed at expected operating pressures for durations well in excess of the estimated burn time, on the order of one or two hours.

Risk Evaluation

Below is an FMEA matrix used during the design review process.

Failure Modes and Effects Analysis				
Component	Failure Modes	Effect	Overall Risk Level	Mitigation/Validation
Motor Casing	1. Burst/Overpressure 2. Burn Through	1a. RUD 1b. Pressure Loss 2a. RUD 2b.		1a. Hydrostatic test at 1.5X MEOP. 1b. Design with a FOS > 3. 2. Design for casing temp. to remain below 200C
Radial Casing Bolts	1. Bolt Shear 2. Vibration Loosening	1. RUD 2. RUD		1. Hydrostatic test at 1.5X MEOP. 2. Additional preload on bolts.
O-Ring	1. Seal Failure	1a. Pressure Loss 1b. Casing Burn-Through		1. Design with seal redundancy.

Fig 6: FMEA Matrix

Theoretical Budget and Cost

The allotted budget for the Thrust Chamber is \$300 USD. This does not include costs associated with other sub-assemblies such as the injector, nozzle, etc. The preliminary BoM for the assembly comes out to a grand total of \$242.65 USD. This cost also includes enough material for the production of two motor casings, giving more room for error during the manufacturing process and bringing the per-unit cost down to approximately \$136.83 USD. This is well below the amount granted by the original budget outline and any leftover funds will be used to support the development of any neighboring subsystems that run over budget.

What's Next

After the manufacturing of the Thrust Chamber and all of its associated components, the setup will undergo hydrostatic testing. Pending a successful test, the system will then perform a cold flow test to validate injector performance before being interfaced with the oxidizer tank and associated plumbing. After all systems are checked and finalized, a static fire will be performed on a test stand and performance data will be gathered to provide correlation between simulated and experimental values.

Appendix A - Reference Papers and Theory

Half-Cat Rocketry Vehicle Design Guide HCR-5100

https://drive.google.com/file/d/1EzwgKEPsJ50NjWJJ_NhrYJfyHFjFNQsr/view

Parker O-Ring Handbook ORD-5700

<https://www.parker.com/content/dam/Parker-com/Literature/O-Ring-Division-Literature/ORD-5700.pdf>

Waterloo Rocketry Vidar III

https://www.waterloorocketry.com/pdfs/2017_project_report.pdf

Appendix B - Calculations

Pressure Calculation Spreadsheet

https://docs.google.com/spreadsheets/d/1Cxr274YuqDjPJZjc4YcJ2lBqTBuR1vpQ/edit?usp=drive_link&oid=108502510649229835822&rtpof=true&sd=true

Bill of Materials

https://docs.google.com/spreadsheets/d/1vQBEQ9TeQU7EqZwEn80_DmLYfDx1Nhcp/edit?usp=drive_link&oid=108502510649229835822&rtpof=true&sd=true